Separate page-addressed apparatus

Authority physical page 1; repository-cover disposition

Authority physical page 1 — repository-cover boundary.

Disposition: excluded from both scholarly article bodies; retained as provenance/control. The page is a vector NUMDAM cover with no article folio and no embedded raster page image. It carries the cover-form title “La conjecture de Weil : I,” the author name PIERRE DELIGNE, journal/volume/year/page-span metadata, and six Link/URI annotations. The exact annotation targets and rectangles are preserved in the structured page record but are deliberately not rendered into either article body.

Excluded copy matter: repository access locator, copyright and conditions text, commercial-use warning, digitization-program notice, and repository branding. The French and English articles therefore begin at authority physical page 2/ printed page 273.

Authority content-stream SHA-256: 1CC909D44EABC4883066D2BC01AADB86E5D7737775448C2492F286B 7CC9F5230. Six annotations were individually inventoried and hashed.

Presentation policy: the French and English readers are separate editable native-math TeX editions. Running heads and repeated folios are excluded from the article bodies and retained only as provenance/page anchors. Source anomalies are identified in this apparatus; no emendation is silently substituted. Synthetic TIFF-header serialization differences across runtimes are separated from reproducible one-bit source-pixel identity in FINAL_PIXEL_PROVENANCE.tsv.

Authority physical page 2; printed 273

Authority physical page 2/ printed page 273.

Included objects: article-form title “LA CONJECTURE DE WEIL. I,” byline “par PIERRE DELIGNE,” the eight-entry Sommaire, three introductory paragraphs, Section 1 heading, paragraph (1.1), display (1.1.1), convergence parenthesis, and the closing sentence introducing the finite-field case. The printed folio 273 and gathering signature 35 are retained as page anchors/provenance, not body text.

Translation choices: “La majoration fondamentale” is rendered “The fundamental estimate”; “théorème de Lefschetz « difficile »” is rendered “the ‘hard’ Lefschetz theorem.” No translator credit, revision note, sequel footnote, repository locator, or other non-source matter from the 31-page retypeset comparator was admitted.

Both 35-page comparators visually align with the authority on this page; the authority remains controlling. No image fallback was required because the title, prose, and display are safely encoded as editable native mathematical TeX.

Authority physical page 3; printed 274

Authority physical page 3/ printed page 274.

Included article objects: continuation of (1.1), displays (1.1.2) and (1.1.3), paragraphs (1.2) and (1.3), and the opening of (1.4). The running author head and repeated printed folio 274 are retained only as page provenance, excluded from the scholarly body.

Authority-controlled readings rejected from inherited TeX: (1.1.2)-(1.1.3) read $\mathrm{Z}(\mathrm{X};t)$ and $\mathrm{Z}(\mathrm{X};q^{-s})$, not $\mathrm{Z}(\mathrm{X}_0;t)$; the last sentence of (1.3) concerns $\mathrm{H}^i_c(\mathrm{X},\mathbf{Q}_\ell)$, not $\mathrm{H}^i(\mathrm{X},\mathbf{Q}_\ell)$; and (1.4) prints $\mathrm{H}^0(\mathrm{U}_0,\mathcal{O})$, not $\mathrm{H}^0(\mathrm{U},\mathcal{O})$. The Dwork/Lubkin citation ends at pp. 105-255; the retyped English comparator's 125-255 was rejected.

The IAS and collected-split scans agree with the authority's text and formulas but do not control readings. No image fallback was required.

Authority physical page 4; printed 275

Authority physical page 4/ printed page 275.

Included article objects: the continuation of (1.4), including items a)-d), the orbit notation $|\mathrm{X}|_{\mathrm{F}}$, and display (1.4.1); paragraph (1.5); the Lefschetz trace formula; displays (1.5.1), (1.5.2), and (1.5.3). Repeated folio 275 is retained as provenance only.

The summation in (1.4.1) is indexed by $\deg(x)|n$, with x understood from the immediately preceding $|\mathrm{X}_0|$ discussion. Formula (1.5.2) preserves the printed double-negative presentation before its positive series expansion. The standalone English follows the frozen French and retains all equation numbers.

Both external comparators align visually; OCR witnesses substantially corrupt symbols, subscripts, superscripts, and ℓ-adic notation and were not imported. No image fallback was required.

Final authority replay: the orbit-set notation is $|\mathrm{X}|_{\mathrm{F}}$, with F a subscript. A quotient slash in the earlier French record was removed.

The last sum in (1.5.2) has subscript n only. The immediately preceding double sum retains $n > 0$; no condition is supplied to the last sum.

Authority physical page 5; printed 276

Authority physical page 5/ printed page 276.

Included article objects: completion of the formal-series argument; display (1.5.4); the explanatory paragraph; Theorem (1.6); Lemma (1.7); and the proof through the definition of $\mathrm{P}_i(t)$. Repeated folio 276 is retained as provenance only.

Critical authority reading: Theorem (1.6) states the characteristic polynomial $\det(t \cdot 1 - \mathrm{F}^*, \mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell))$. The inherited French TeX and the 31-page English retypeset change this to $\det(1 - \mathrm{F}^* t, \mathrm{H}^i)$, whose roots are reciprocals; that substitution was rejected. By contrast, $\mathrm{P}_i(t) = \det(1 - \mathrm{F}^* t, \mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell))$ in the proof is retained exactly because that is what the authority prints there.

The inclusion $\mathrm{Z}(\mathrm{X}_0, t) \in \mathbf{Z}[[t]] \cap \mathbf{Q}_\ell(t) \subset \mathbf{Q}(t)$, the Hankel determinant indices, and the exponent $(-1)^{i+1}$ in (1.5.4) were checked against authority pixels. No image fallback was required.

Authority physical page 6; printed 277

Authority physical page 6/ printed page 277.

Included article objects: the proof's products $\mathrm{P}(t)$ and $\mathrm{Q}(t)$; the argument using $\mathrm{K}, \mathrm{R}(t)$, Gauss's lemma, and roots of absolute value $q^{-i/2}$; paragraphs (1.8), (1.9)$(a) - (c)$, and (1.10), including the category equivalence and tensor-product display. Repeated folio 277 is retained as provenance only.

Translation terminology: "$\mathbf{Q}_\ell$-faisceau constant tordu" is rendered "twisted-constant $\mathbf{Q}_\ell$-sheaf" to preserve the source's technical distinction rather than silently replacing it with a modernized term. $\mathrm{X}^{\mathrm{an}}, \pi_1(\mathrm{X}, \overline{x}), \mathbf{Z}_\ell, \mathbf{Q}_\ell$, support-proper cohomology H^i_c, and all subscripts/superscripts were checked against the authority.

The retyped comparator is useful for orientation but simplifies "constant tordu" to "locally constant" and adds non-source matter elsewhere; it was not accepted as controlling translation. No image fallback was required.

Final authority replay: the category map is rightward mapsto, printed again at the wrapped line. The repeated glyph indicates the same continued map, not a reverse equivalence arrow.

Authority physical page 7; printed 278

Authority physical page 7/ printed page 278.

Included article objects: the end of (1.10); paragraph (1.11), including the étalé-space construction and its commutative square; paragraph (1.12), including the two-stage cohomology map, the Lefschetz trace formula, and (1.12.1); and paragraph (1.13), including the local-factor definition. The repeated folio 278 is retained only as page provenance.

Critical authority readings: both horizontal arrows in the (1.11) square are labelled F, and the ensuing fibre product is $\mathrm{X} \times_{\mathrm{F},\mathrm{X},f} [\mathcal{F}] = [\mathrm{F}^*\mathcal{F}]$. Formula (1.12) visibly passes through $\mathrm{H}^i_c(\mathrm{X}, \mathrm{F}^*\mathcal{F})$ before $\mathrm{H}^i_c(\mathrm{X}, \mathcal{F})$. In (1.13), the local operator is $\mathrm{F}^*_{x_0} = \mathrm{F}_x^{*\,\deg(x_0)}$; the subscript, star, and degree of x_0 are all retained. The inherited Paper_20.tex simplifies or changes each of these points and was rejected as an automatic source.

The square is simple enough to preserve safely as aligned editable text; no raster fallback was required. IAS and collected-split comparator pages 6 visually align with the authority but do not control readings.

The two canonical sheaf isomorphisms retain the source rightward arrows with a tilde above.

Authority physical page 8; printed 279

Authority physical page 8/ printed page 279.

Included article objects: (1.14), with formulas (1.14.1)-(1.14.3); and the complete Galois dictionary (1.15), including (1.15.1), (1.15.2), and the Galois form of (1.14.3). The repeated folio 279 is retained only as page provenance.

Critical authority readings: (1.14.1) retains F_x^* and $t^{\deg(x)}$; (1.14.2) retains the displayed defining quotient and the double sum over n and $x \in \mathrm{X}^{\mathrm{F}^n} = \mathrm{X}_0(\mathbf{F}_{q^n})$; (1.15.2) reads $\mathrm{F}_{x_0}^* \underset{\text{dfn}}{=} \mathrm{F}_x^{*\,\deg(x_0)} = \mathrm{F}_{x_0}$. In the final Galois formula, the F_x and F are unstarred, exactly as printed. No inherited normalization of these distinctions was admitted.

All displays were encoded as editable linear mathematics. Both external comparator pages 7 align visually; OCR substantially damages stars, exponents, stalk notation, and summation alignment and was used only as a locator.

Both the field isomorphism and induced cohomology isomorphism retain their source rightward arrows with a tilde above.

The equality annotations in (1.14.2) and (1.15.2) preserve the printed abbreviation dfn, including in the English reader; they are centered below the equality sign.

Authority physical page 9; printed 280

Authority physical page 9/ printed page 280.

Included article objects: the Section 2 heading; $(2.1)(a) - (b)$, including all three orientation choices and the definitions of $\mathbf{Z}(1)$ and $\mathbf{Z}(r)$; and (2.2) through the projective system and negative Tate twist. The repeated folio 280 is retained only as page provenance.

Critical authority readings: for an n-dimensional differentiable manifold the fundamental class is $\mathrm{Tr} : \mathrm{H}_c^n(\mathrm{X}, \mathbf{Z}') \longrightarrow \mathbf{Z}$, not H_c^{2n}. The acting group in (2.2) is the unit group $\hat{\mathbf{Z}}^*$ of the profinite completion, and the transition maps are the full family $\sigma_{m,n} : \mathbf{Z}/\ell^m(1) \longrightarrow \mathbf{Z}/\ell^n(1), x \longmapsto x^{\ell^{m-n}}$. For negative r, $\mathbf{Q}_\ell(r) = \mathbf{Q}_\ell(-r)^\vee$. The inherited French TeX changes or suppresses these readings and was not accepted.

Translation choices: “langage farfelu” is rendered “fanciful language,” “variétés différentiables” as “differentiable manifolds,” and the source’s notation and nested a)-c) structure are retained. No image fallback was required.

Final authority replay restores the printed dual superscript vee in the negative Tate twist.

Authority physical page 10; printed 281

Authority physical page 10/ printed page 281.

Included article objects: the continuation of (2.2), including the Frobenius action and the two fundamental-class maps; Theorem (2.3); paragraphs (2.4)-(2.5); and the opening functional equation in (2.6). The folio 281 and gathering signature 36 are retained as provenance, not body text.

Critical authority readings: Theorem (2.3) displays $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell) \otimes \mathrm{H}^{2n-i}(\mathrm{X}, \mathbf{Q}_\ell) \longrightarrow \mathbf{Q}_\ell(-n)$; the second displayed factor is not written with an (n)-twist, although the parenthetical identification refers to the dual of $\mathrm{H}^{2n-i}(\mathrm{X}, \mathbf{Q}_\ell(n))$. The functional equation is

$\mathrm{Z}(\mathrm{X}_0, t) = \varepsilon \cdot q^{\frac{-n\chi(\mathrm{X})}{2}} \cdot t^{-\chi(\mathrm{X})} \cdot \mathrm{Z}(\mathrm{X}_0, q^{-n} t^{-1})$.

The negative exponents of q and t were checked directly against the authority. Paper_20.tex reverses them and was rejected. No image fallback was required.

Authority physical page 11; printed 282

Authority physical page 11/ printed page 282.

Included article objects: the completion of (2.6), including the definition of N and the piecewise value of ε; (2.7); Theorem (2.8) with its full chain of pairings; (2.9), including (2.9.1) and the invariant/coinvariant identities; Scholium (2.10); and the opening of (2.11). The repeated folio 282 is retained only as page provenance.

Critical authority readings: N is introduced when n is even; $\varepsilon = 1$ for n odd and $\varepsilon = (-1)^{\mathrm{N}}$ for n even. The dual sheaf is $\mathcal{F}^\vee$, and the full (2.8) chain through $\mathrm{H}_c^{2n}(\mathrm{X}, \mathcal{F} \otimes \mathcal{F}^\vee(n))$ and $\mathrm{H}_c^{2n}(\mathrm{X}, \mathbf{Q}_\ell(n))$ is retained. The direct image beginning (2.11) is $j_*\mathcal{F}$, not $j_!\mathcal{F}$. The inherited French TeX says "n impair" at the N-definition and substitutes $j_!$, so it remains a divergence witness only.

"Constant tordu" continues to be rendered "twisted-constant," preserving the source's technical distinction. No image fallback was required.

Final authority replay restores all seven printed dual superscript vees and the rightward isomorphism arrow in (2.9.1).

Authority physical page 12; printed 283

Authority physical page 12/ printed page 283.

Included article objects: the completion of (2.11); Theorem (2.12); (2.13); all five bibliographical notes in (2.14); the Section 3 heading; the Rankin sentence; and the opening of (3.1). The repeated folio 283 is retained only as page provenance.

Critical authority readings: every direct-image occurrence in (2.12) is j_*; the pairing runs through ordinary H^2, not H^2_c, and ends with $\mathrm{H}^2(\mathrm{X}, j_*\mathbf{Q}_\ell(1)) = \mathrm{H}^2(\mathrm{X}, \mathbf{Q}_\ell(1)) \longrightarrow \mathbf{Q}_\ell$. Bibliographical note E cites SGA 4, XVIII (3.2.5), with $\mathrm{K} = j_*\mathcal{F}$ and $\mathrm{L} = \mathbf{Q}_\ell$. Paper_20.tex instead uses $j_!$ or a mixed $j_!/j_*$ formula and cites (3.2.4.1); the 31-page retypeset also changes the H^2 step. Those readings were rejected.

"La majoration fondamentale" is rendered consistently as "The fundamental estimate," rather than the comparator's "The main lemma." All mathematics and the section transition were safely encoded as editable text; no image fallback was required.

Final authority replay restores the three printed dual superscript vees in the chain of pairings (2.12).

Authority physical page 13; printed 284

Authority physical page 13/ printed page 284.

Included article objects: the definition of weight β; Theorem (3.2), including its three hypotheses; Lemmas (3.3)-(3.6); every displayed tensor-power, trace, determinant, coefficient, and radius-of-convergence formula. The repeated folio 284 is retained only as page provenance.

Critical authority readings: the page first allows $\beta \in \mathbf{Q}$, while hypothesis (3.2)(i) prints the pairing $\psi : \mathcal{F}_0 \otimes \mathcal{F}_0 \longrightarrow \mathbf{Q}_\ell(-\beta)$ with the explicit parenthetical qualification ($\beta \in \mathbf{Z}$). In (3.3), the local determinant contains $t^{\deg(x)}$, and the trace on the $(2k)$-fold tensor power is $\mathrm{Tr}(\mathrm{F}_x^n, \mathcal{F}_0)^{2k}$. Lemma (3.6) compares the infima of the moduli of poles with the printed $\leq$ direction. These distinctions were checked directly against the authority pixels.

The inherited French TeX was consulted only after the diplomatic record was frozen. It omits or regularizes some of these qualifications and was not accepted automatically. All formulas were safely preserved as editable linear mathematics; no image fallback was required.

Translation: "positifs" in (3.3)-(3.5) is rendered "nonnegative"; the proof explicitly gives ≥ 0 and an even trace power can vanish. This is the source meaning, not a strict-positivity assertion.

Authority physical page 14; printed 285

Authority physical page 14/ printed page 285.

Included article objects: paragraph (3.7), including the pair partitions and contractions ψ_{P}; the Weyl coinvariant statement; the compact-support cohomology calculation; the reduced Z-function; and the complete upper- and lower-bound argument for $|\alpha|$. The repeated folio 285 is retained only as page provenance.

Critical authority readings: the source speaks of coinvariants, not invariants, and gives

$(\overset{2k}{\otimes}\mathcal{F}_u)_{\pi_1} = (\overset{2k}{\otimes}\mathcal{F}_u)_{\mathrm{Sp}} \simeq \mathbf{Q}_\ell(-k\beta)^{\mathcal{P}'}, \mathrm{H}^2_c(\mathrm{U}, \overset{2k}{\otimes}\mathcal{F}) \simeq \mathbf{Q}_\ell(-k\beta-1)^{\mathrm{N}}, \mathrm{H}^0_c(\mathrm{U}, \overset{2k}{\otimes}\mathcal{F}) = 0.$

The numerator of the displayed Z-function is ordinary $\mathrm{H}^1(\mathrm{U}, \overset{2k}{\otimes}\mathcal{F})$, while the denominator is $(1-q^{k\beta+1}t)^{\mathrm{N}}$. The inverse local pole is $1/\alpha^{2k/\deg(x)}$; the authority then derives $|\alpha| \leq q_x^{\frac{\beta}{2}+\frac{1}{2k}}$, lets k tend to infinity, and uses the paired eigenvalue $q_x^{\beta}\alpha^{-1}$ for the reverse inequality.

Paper_20.tex changes the cohomological degrees/supports, the pole, the exponent of α, and the inequality chain; those readings were rejected. No image fallback was required.

The coinvariant comparison uses a rightward isomorphism arrow; the following H^2_c comparison has the undirected sign $\simeq$. These distinct source forms are retained.

Authority physical page 15; printed 286

Authority physical page 15/ printed page 286.

Included article objects: Corollary (3.8) and its proof, including the Euler-product convergence estimate; Corollary (3.9); the short exact sequence relating $j_!\mathcal{F}$ and $j_*\mathcal{F}$; and the resulting cohomology segment. The repeated folio 286 is retained only as page provenance.

Critical authority readings: Corollary (3.8) gives $|\alpha| \leq q^{\frac{\beta+1}{2}+\frac{1}{2}}$, and the defining Euler product is shown to converge absolutely for $|t| < q^{\frac{-\beta}{2}-1}$. The degree-$n$ point count is bounded by q^n and yields $\sum_n q^{-n\varepsilon} < \infty$. Corollary (3.9) states
$q^{\frac{\beta+1}{2}-\frac{1}{2}} \leq |\alpha| \leq q^{\frac{\beta+1}{2}+\frac{1}{2}}$.
The exact sheaf sequence is $0 \longrightarrow j_!\mathcal{F} \longrightarrow j_*\mathcal{F} \longrightarrow j_*\mathcal{F}/j_!\mathcal{F} \longrightarrow 0$, followed by $\mathrm{H}^1_c(\mathrm{U},\mathcal{F}) \longrightarrow \mathrm{H}^1(\mathbf{P}^1, j_*\mathcal{F}) \longrightarrow 0$. The inherited TeX changes both the bounds and the sequence, so it remains a divergence witness only. No image fallback was required.

Authority physical page 16; printed 287

Authority physical page 16/ printed page 287.

Included article objects: completion of Corollary (3.9); the Section 4 heading; all hypotheses and constructions in (4.1); the specialization and monodromy descriptions; the vanishing-cycle exact sequence; the Picard-Lefschetz formula; and the complete four-column sign table. The repeated folio 287 is retained only as page provenance.

Critical authority readings: Poincaré duality supplies the eigenvalue $q^{\beta+1}\alpha^{-1}$. In the exact sequence for $i = n, n+1$, the middle map is $x \longmapsto (x,\delta)$. The table reads, for n mod 4 equal to 0,1,2,3 respectively:

sign in Tx=$x \pm (x,\delta)\delta : -,-,+,+$; $(\delta,\delta) : 2,0,-2,0$; T$\delta : -\delta,\delta,-\delta,\delta$.

Both the inherited French TeX and the retypeset comparator contain sign-table errors; the authority table controls. The table and exact sequence were safely encoded as editable mathematical arrays; no raster fallback was required.

The first specialization composite has a leftward isomorphism arrow; the later comparison to the generic fiber has a rightward isomorphism arrow. Neither is an undirected congruence sign.

The map $x \longmapsto (x,\delta)$ is printed above its exact-sequence arrow; the editable encoding records an above-arrow label.

Authority physical page 17; printed 288

Authority physical page 17/ printed page 288.

Included article objects: the intersection-form statement; all of (4.2), including the henselian trait, specialization map (4.2.1), and inertia action (4.2.2); and all of (4.3) through the odd-dimensional local monodromy formula. The repeated folio 288 is retained only as page provenance.

Critical authority readings: the geometric generic point is $\overline{\eta}$ and $\mathrm{I} = \mathrm{Gal}(\overline{\eta}/\eta)$. Formula (4.2.1) passes through $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell)$. In (4.3.1), δ belongs to $\mathrm{H}^n(\mathrm{X}_{\overline{\eta}}, \mathbf{Q}_\ell)(m)$. The exact sequence (4.3.3) carries the map $x \longmapsto \mathrm{Tr}(x \cup \delta)$ into $\mathbf{Q}_\ell(m-n)$, and the odd-dimensional action uses the canonical homomorphism $t_\ell : \mathrm{I} \longrightarrow \mathbf{Z}_\ell(1) : x \longmapsto x \pm t_\ell(\sigma)(x, \delta)\delta$.

A faint vertical line in the lower half of the scan is a source-image artifact rather than article content. It is retained in the hashed pixel replay but excluded from the diplomatic and English text. No image fallback was required.

In (4.2.1) the first isomorphism points left; (4.3.2) points right. Both source directions are preserved.

In (4.3.3), $x \longmapsto \mathrm{Tr}(x \cup \delta)$ is printed above the arrow, not below it.

Authority physical page 18; printed 289

Authority physical page 18/ printed page 289.

Included article objects: the even-dimensional case B); all of (4.4); the Section 5 heading; the complete opening of (5.1); and projection diagram (5.1.1), including both arrow labels and the sentence identifying the fiber. The folio 289 and gathering signature 37 are retained as provenance, not body text.

Critical authority readings: for $p \neq 2$ the character is $\varepsilon : \mathrm{I} \longrightarrow \{\pm 1\}$; the second action line applies when $\varepsilon(\sigma) = -1$. In (4.4)(a), $\mathrm{R}^n f_* \mathbf{Q}_\ell = j_* j^* \mathrm{R}^n f_* \mathbf{Q}_\ell$. In the exceptional $\delta = 0$ case, the exact sequence begins with the sheaf $\mathbf{Q}_\ell(m-n)_s$ and ends with $j_* j^* \mathrm{R}^{n+1} f_* \mathbf{Q}_\ell$. Section (5.1) prints $\dim(\mathbf{P}) \geq 1$ and uses the dual projective space $\check{\mathbf{P}}$.

Diagram (5.1.1) has $\widetilde{\mathrm{X}}$ mapping left to X by π and down to D by f. Direct authority inspection established that this simple topology could be preserved safely in editable mathematical arrays. The inherited TeX instead substitutes a different square, and its reading was rejected. No untouched-crop/conservative-derivative fallback pair was required.

Authority physical page 19; printed 290

Authority physical page 19/ printed page 290.

Included: the completion of the genericity conditions in (5.1); all of (5.2); the hand-drawn loops figure; formula (5.2.1); Proposition (5.3); Theorem (5.4); and the opening sentence on the dual variety. The running author head is retained only as provenance, excluded from the scholarly body; the printed folio is a page anchor, not body text.

Canonical transcription notes: both occurrences in the transcendental discussion use ordinary $\mathbf{Q}$, not $\mathbf{Q}_\ell$. The page ends at the printed hyphen in « irré- », continued on the next page. The loops figure is not safely reducible to linear text: asset $\mathbf{P}0019-\mathrm{A}01$ preserves an untouched one-bit crop from the authority image and a separately named conservative presentation derivative. Exact pixel bounds, operations, and hashes are in the asset ledger.

Authority physical page 20; printed 291

Authority physical page 20/ printed page 291.

Included: the completion of « irréductible »; the incidence diagram $X \longleftarrow Y \longrightarrow \check{\mathbf{P}}$; the fundamental-group surjection; the proof of Theorem (5.4); Corollary (5.5); and the opening of the abstract algebraic version in (5.6). The folio is retained only as provenance.

Canonical transcription notes: the proof takes $F \subset E \otimes \mathbf{C}$ and assumes $F \not\subset (E \cap E^{\perp}) \otimes \mathbf{C}$. The projective space in (5.6) has dimension >1. The simple incidence diagram is preserved as editable mathematical arrays with g on the downward arrow. A different inherited square was not used.

Authority physical page 21; printed 292

Authority physical page 21/ printed page 292.

Included: the completion of (5.6); all of (5.7); and the nonzero-vanishing-cycle case of (5.8), including every sheaf index, invariant subspace, and monodromy generator.

Canonical transcription notes: the Veronese embedding is $\iota_{(r)}$, the displayed dimension is $\binom{N+r}{N} - 1$, the given embedding is ι_1, and $\iota_r = \iota_{(r)} \circ \iota_1$. In (5.8), the excluded case is $p = 2$ with n even; the ramification statement is for $R^n f_* \mathbf{Q}_\ell$, while the later control along the dual variety is for $R^i g_* \mathbf{Q}_\ell$.

The Veronese subscript is literally (r), whereas the later composite is denoted with plain r. Braces in the machine-readable encoding disambiguate those literal parentheses.

Authority physical page 22; printed 293

Authority physical page 22/ printed page 293.

Included: the zero-vanishing-cycle case of (5.8); (5.9); Theorem (5.10); and Lemma (5.11) through part b). The folio is not body text.

Canonical transcription notes: the Lie algebra is represented consistently by the Fraktur capital L, $\mathfrak{L}$. Its generators are $\mathrm{N}_s : x \longmapsto (x, \delta_s)\delta_s$; Lemma (5.11)(ii) has $x \longmapsto \psi(x, \delta)\delta$, without an inserted $\pm$ sign. The exact sequence in the zero-cycle case begins with $\bigoplus_{s\in S} \mathbf{Q}_\ell(m-n)_s$.

Authority physical page 23; printed 294

Authority physical page 23/ printed page 294.

Included: the conclusion of Lemma (5.11); Remark (5.12); the bibliographical indications (5.13); the heading of §6; and the opening of (6.1).

Canonical transcription notes: the maximal subspaces are W_α, hypothesis (i) gives $\mathrm{W}_\alpha = \mathrm{V}$, and $\mathfrak{L}$ contains every $\mathrm{N}(\delta)$. In Remark (5.12), the denominator is $\prod_{i=0}^{n}(1-q^i t)$. The projective space $\mathbf{P}_0$ at the opening of §6 has dimension ≥ 1.

Authority physical page 24; printed 295

Authority physical page 24/ printed page 295.

Included: diagram (6.1.1); the full setup of (6.1); Theorem (6.2); Corollary (6.3); and Lemma (6.4). The folio is retained as provenance.

Canonical transcription notes: diagram (6.1.1) has $\widetilde{\mathrm{X}}_0$ mapping left to X_0 by π_0 and down to D_0 by f_0. Theorem (6.2) concerns $\det(1-\mathrm{F}_x^*t,\mathcal{E}_0/(\mathcal{E}_0\cap\mathcal{E}_0^{\perp}))$. The bounds in Corollary (6.3) are $q^{\frac{n+1}{2}-\frac{1}{2}}\leq|\alpha|\leq q^{\frac{n+1}{2}+\frac{1}{2}}$. Lemma (6.4) ends with $\alpha_i^{\deg(x)}$.

The source literally prints $\mathrm{R}^i f_*\mathbf{Q}_\ell$ and $\mathrm{R}^i f_{0*}\mathbf{Q}_\ell$ in the sentence beginning “Ce dernier est défini”. These two i indices are retained; adjacent occurrences of n are not silently substituted for them.

Authority physical page 25; printed 296

Authority physical page 25/ printed page 296.

Included: completion of Lemma (6.4); the factorization of $\mathrm{Z}(\mathrm{X}_x, t)$ into Z^f and Z^m; the definitions of $\mathcal{F}_0$ and $\mathcal{F}$; paragraph (6.5); and the opening of Proposition (6.6). The printed folio is recorded only as a page anchor.

Canonical transcription notes: the displayed factorization uses Z^f and Z^m. The formula after the definitions of $\mathcal{F}_0$ and $\mathcal{F}$ has $\prod_j (1 - \beta_j^{\deg(x)} t)$ in the denominator and $\det(1 - \mathrm{F}_x^* t, \mathcal{F}_0)$ as a multiplicative factor. In (6.6.1), $\det(1 - \mathrm{F}_x^* t, \mathcal{F}_0)$ occurs in the numerator.

Authority physical page 26; printed 297

Authority physical page 26/ printed page 297.

Included: Lemma (6.7), Proposition (6.8), paragraphs (6.9) and (6.10), and every displayed polynomial and divisibility statement. The printed folio and signature mark are recorded only as page anchors.

Canonical transcription notes: the authority literally prints “unités p-adiques dans $\overline{\mathbf{Q}}_\ell$” in Proposition (6.8); the French record retains that wording, and the English translates it as “p-adic units in $\overline{\mathbf{Q}}_\ell$,” without silent emendation. $\mathrm{R}(t)$ is characterized as the least common multiple of the polynomials $\mathrm{S}(t)$. The polynomials used in (6.9) are $\prod_i(1-\alpha_i^{\deg(x)}t)\cdot\det(1-\mathrm{F}_x^*t,\mathcal{F}_0)$. Paragraph (6.10) uses the fiber $\mathcal{F}_u$ and the arithmetic group $\hat{\mathbf{Z}}=\mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$.

The divisibility hypothesis in (6.8) literally has product index i and δ_i. This printed choice is retained, alongside the surrounding δ_j family.

The printer signature 38 below the lower folio 297 is retained in source-image provenance and explicitly inventoried as P0026 $-$ $\mathcal{O}$08; it is excluded from both scholarly bodies.

Authority physical page 27; printed 298

Authority physical page 27/ printed page 298.

Included: the definition of H and ρ_1; Lemmas (6.11) and (6.12); paragraph (6.13); the heading of §7; and Lemma (7.1). The printed folio is recorded only as a page anchor.

Canonical transcription notes: $\mathrm{H} \subset \hat{\mathbf{Z}} \times \mathrm{CSp}(\mathcal{F}_u, \psi)$ is defined by $q^{-n} = \mu(g)$, with $n \in \hat{\mathbf{Z}}$. Lemma (6.12) uses an ℓ-adic unit δ and the fibers Z_n. Paragraph (6.13) uses $\beta_j^{\deg(x)}$. The bounds in (7.1.1) are $q^{\frac{d}{2}-\frac{1}{2}} \leq |\alpha| \leq q^{\frac{d}{2}+\frac{1}{2}}$.

In the Chebotarev sentence of (6.13), the source prints $\beta_j^{\deg(x)}$, not $\delta_j^{\deg(x)}$. The same source reading is now retained in English.

Authority physical page 28; printed 299

Authority physical page 28/ printed page 299.

Included: the scalar-extension reduction in Lemma (7.1); assumptions a)-d); the blow-up cohomology statement; the Leray spectral sequence; and cases A) and B) through the page boundary. The printed folio is recorded only as a page anchor.

Canonical transcription notes: the authority gives an injection $\mathrm{H}^i(\mathrm{X}, \mathbf{Q}_\ell) \hookrightarrow \mathrm{H}^i(\widetilde{\mathrm{X}}, \mathbf{Q}_\ell)$ and then the direct-sum decomposition. The spectral sequence is $\mathrm{E}_2^{pq} = \mathrm{H}^p(\mathrm{D}, \mathrm{R}^q f_* \mathbf{Q}_\ell) \Rightarrow \mathrm{H}^{p+q}(\widetilde{\mathrm{X}}, \mathbf{Q}_\ell)$. Case A) is $\mathrm{E}_2^{2,n-1}$, and case B) is $\mathrm{E}_2^{0,n+1}$.

The source Leray superscript is pq without an inserted comma. The two projective-embedding arrows in the prose are ordinary rightward arrows; the weak Lefschetz comparison in case A is an injection, not an asserted isomorphism.

The three extension-field labels in the first paragraph are $\mathbf{F}_{q'}$, as printed. The source subsequently states that q is replaced by q^r; that separate power is unchanged.

Authority physical page 29; printed 300

Authority physical page 29/ printed page 300.

Included: completion of case B); case **C**); exact sequences (7.1.2)-(7.1.5) and their cohomological consequences; and Lemma (7.2). The printed folio is recorded only as a page anchor.

Canonical transcription notes: $\mathrm{R}^n f_* \mathbf{Q}_\ell$ is filtered by $j_*\mathcal{E}$ and $j_*(\mathcal{E} \cap \mathcal{E}^\perp)$. In (7.1.3), the constant sheaf is $j_*(\mathcal{E} \cap \mathcal{E}^\perp)$. In (7.1.4), the constant subsheaf is $j_*\mathcal{E}^\perp$. Frobenius acts on $\mathbf{Q}_\ell(n-m)$ by $q^{d/2}$, and Lemma (7.2) states $|\alpha| = q^{d/2}$.

Authority physical page 30; printed 301

Authority physical page 30/ printed page 301.

Included: the reduction of Lemma (7.2) to Lemma (1.7); paragraph (7.3); the heading of §8; Theorem (8.1); and the opening of its proof. The printed folio is recorded only as a page anchor.

Canonical transcription notes: the reduction passes through $0 \leq i \leq n$. In (7.3), the bounds are $q^{\frac{kd}{2}-\frac{1}{2}} \leq |\alpha^k| \leq q^{\frac{kd}{2}-\frac{1}{2}}$, hence $q^{\frac{d}{2}-\frac{1}{2k}} \leq |\alpha| \leq q^{\frac{d}{2}+\frac{1}{2k}}$. Theorem (8.1) gives $|\#\mathrm{X}_0(\mathbf{F}_q) - \#\mathbf{P}^n(\mathbf{F}_q)| \leq b \cdot q^{n/2}$, and its proof opens with the line $\mathbf{Q}_\ell \cdot \eta^i \subset \mathrm{H}^{2i}(\mathrm{X}, \mathbf{Q}_\ell)$.

The first displayed inequality of (7.3) literally has kd/2 $-\ \frac{1}{2}$ in both exponents. The next display has upper exponent $\frac{d}{2} + \frac{1}{2k}$. This source anomaly is preserved rather than silently repaired. The source also retains the α component index in c) and $\mathbf{P}_0$ in the ambient space of (8.1).

Authority physical page 31; printed 302

Authority physical page 31/ printed page 302.

Included: the conclusion of Theorem (8.1), Theorem (8.2), Remark (8.3), and the opening hypotheses of Theorem (8.4). The running author head and both appearances of folio 302 are excluded from article text and retained only as provenance/page anchors.

Canonical notes: the modular transformation factor is $\varepsilon(a)^{-1}$; the normalized q-expansion begins with $q = e^{2\pi i z}$; and the source spells Miyake without an accent. The Weil bound is retained as $|a_p| \leq 2 \cdot p^{\frac{k-1}{2}}$, followed by $\mathrm{T}^2 - a_p\mathrm{T} + \varepsilon(p) \cdot p^{k-1}$. Comparator and prior-work readings were consulted only after the immutable $v1$ freeze; all inherited members remain ZERO_ACCEPTED.

The long dash separating each numbered theorem, lemma, or remark label from its statement is retained from the authority; it is not replaced by a hyphen.

Authority physical page 32; printed 303

Authority physical page 32/ printed page 303.

Included: the exponential-sum bound in Theorem (8.4); equations (8.4.1)-(8.4.3); the Artin-Schreier cover, Frobenius calculation, and rank-one λ-adic systems; and Lemma (8.5). The running title and both appearances of folio 303 are excluded from article text and retained only as provenance/page anchors.

Canonical notes: the map is $\sigma : \mathrm{X}_0 \longrightarrow \mathrm{A}_0$; the local system is $\mathcal{F}_{j,0}$ with $\iota(i * x) = \psi(-ji)\iota(x)$; and (8.4.3) decomposes $\mathrm{H}^*_c(\mathrm{X}, \mathbf{Q}_\ell) \otimes \mathrm{E}_\lambda$ as $\bigoplus_j \mathrm{H}^*_c(\mathrm{A}, \mathcal{F}_j)$. The perfect pairing retains its trace-labelled final arrow. All inherited prior-work members remain ZERO_ACCEPTED.

The long dash separating each numbered theorem, lemma, or remark label from its statement is retained from the authority; it is not replaced by a hyphen.

Authority physical page 33; printed 304

Authority physical page 33/ printed page 304.

Included: the deduction of (8.4) from (8.5), the trace identity, diagram (8.6.1), Lemmas (8.7) and (8.8), equations (8.7.1), and paragraph (8.9). The running author head and both appearances of folio 304 are excluded from article text and retained only as provenance/page anchors.

Canonical notes: the open immersions remain hooks, $j_0 : \mathrm{X}_0 \hookrightarrow \mathrm{Z}_0$ and $j : \mathrm{X} \hookrightarrow \mathrm{Z}$. Diagram (8.6.1) preserves $\mathrm{X}_0 \hookrightarrow \mathrm{Y}_0$ above $\mathrm{A}_0 \hookrightarrow \mathrm{P}_0 \hookleftarrow \mathrm{P}_0^\infty \hookleftarrow \mathrm{H}_0$, with vertical map σ. The trace expression retains $j_!\sigma^*$. No bitmap fallback is required because the exact topology is safely editable. All inherited prior-work members remain ZERO_ACCEPTED.

The long dash separating each numbered theorem, lemma, or remark label from its statement is retained from the authority; it is not replaced by a hyphen.

Authority physical page 34; printed 305

Authority physical page 34/ printed page 305.

Included: the conclusion of paragraph (8.9), paragraph (8.10), the universal family, the relative compactification square, the derived direct-image decomposition, and the opening of the separation-of-variables argument. The running title and both appearances of folio 305 are excluded from article text and retained only as provenance/page anchors. The lower printer signature 39 is excluded from article text and retained as provenance.

Canonical notes: the products are $\mathrm{A}^n \times \mathrm{S}$ and $\mathbf{P}^n \times \mathrm{S}$. The square preserves $\mathrm{X}_\mathrm{S} \xhookrightarrow{u} \mathrm{Z}_\mathrm{S}$ over $\mathrm{A}_\mathrm{S} \xrightarrow{a} \mathrm{S}$ with right vertical map f, and identifies u as an open immersion. The displayed decomposition is $\mathrm{R}^*(fu)_!(\mathrm{E}_\lambda) = \bigoplus_j \mathrm{R}^* a_! \mathcal{F}_{j,S}$.

No bitmap fallback is required because the exact topology is safely editable. All inherited prior-work members remain ZERO_ACCEPTED.

Authority physical page 35; printed 306

Authority physical page 35/ printed page 306.

Included: the Kunneth reduction, paragraph (8.11), Lemmas (8.12) and (8.13), the Serre reference, the bibliography heading, and entries [1]-[3]. The running author head and both appearances of folio 306 are excluded from article text and retained only as provenance/page anchors.

Canonical notes: the displayed Kunneth formula retains superscript stars in both factors, $\mathrm{H}^*(\mathrm{A}, \mathcal{F}_j) = \otimes \mathrm{H}^*(\mathrm{A}^1, \mathcal{F}_j^1)$. Paragraph (8.11) has $i \neq 1$ and $j \neq 0$, and the extension identity is $u_!\mathcal{F}_j = u_*\mathcal{F}_j$. The two conductor statements and $\chi(\sigma) = \sigma\mathrm{T} - \mathrm{T}$ are retained. All inherited prior-work members remain ZERO_ACCEPTED.

The cohomological comparison in (8.11) retains its rightward isomorphism arrow and tilde.

The long dash separating each numbered theorem, lemma, or remark label from its statement is retained from the authority; it is not replaced by a hyphen.

Authority physical page 36; printed 307

Authority physical page 36/ printed page 307.

Included: bibliography entry [4], the SGA siglum and entries for SGA 4, SGA 5, and both parts of SGA 7, and the manuscript-received date. The running title and both appearances of folio 307 are excluded from article text and retained only as provenance/page anchors. The terminal horizontal rule is excluded from article text and retained as a terminal provenance object.

Canonical notes: the work ends with “Manuscrit reçu le 20 septembre 1973.” There is no author-address block after it. The terminal rule marks the end of the scanned article but contributes no textual content. All inherited prior-work members remain ZERO_ACCEPTED.